

Multiple Choice (10 marks)

1. Statement 1| The Stanford Sentiment Treebank contained movie reviews, not book reviews. Statement 2| The Penn Treebank has been used for language modeling.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. Adding more basis functions in a linear model, pick the most probably option:
 - (a) Decreases model bias
 - (b) Decreases estimation bias
 - (c) Decreases variance
 - (d) Doesn't affect bias and variance
3. Statement 1| Word 2 Vec parameters were not initialized using a Restricted Boltzman Machine. Statement 2| The tanh function is a nonlinear activation function.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. Neural networks:
 - (a) Optimize a convex objective function
 - (b) Can only be trained with stochastic gradient descent
 - (c) Can use a mix of different activation functions
 - (d) None of the above
5. The model obtained by applying linear regression on the identified subset of features may differ from the model obtained at the end of the process of identifying the subset during
 - (a) Best-subset selection
 - (b) Forward stepwise selection
 - (c) Forward stage wise selection
 - (d) All of the above

True / False (10 marks)

Answer True or False

6. True or False: Linear hard-margin SVM can only be effectively used when the training data is linearly separable.
7. True or False: Pruning a Decision Tree is primarily done to avoid overfitting the training set and improve model generalization.
8. True or False: Transforming the data to a zero median is sufficient to obtain the same projection in PCA as in SVD.
9. True or False: Residual connections are commonly utilized in architectures such as ResNets and Transformers to facilitate training deep networks.
10. True or False: K-fold cross-validation exhibits a quadratic complexity as the number of folds, K , increases.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the fundamental differences between Bayesian and frequentist approaches in machine learning, focusing specifically on the role of prior distributions in probabilistic modeling.
12. Discuss the necessary components required to calculate $P(H|E, F)$ without any conditional independence information, and analyze how each component contributes to the calculation.

End of Paper